

SHETH L.U.J. AND SIR M.V. COLLEGE
PRACTICAL NO.:-3
SUB:-Cyber & Information Security

AIM:-Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity.

INPUT:-

```
import hmac
import hashlib

secret_key = b"mysecretkey"

message = input("Enter the message: ").encode()

mac = hmac.new(secret_key, message, hashlib.sha256).hexdigest()

print("\nGenerated MAC (HMAC-SHA256):")
print(mac)

print("\n----- Verify Message -----")
received_message = input("Enter the received message: ").encode()
received_mac = input("Enter the received MAC: ")

new_mac = hmac.new(secret_key, received_message,
hashlib.sha256).hexdigest()

if hmac.compare_digest(received_mac, new_mac):
    print("\nMessage Verified Successfully!")
    print("Integrity and Authenticity are Maintained.")
else:
    print("\nVerification Failed!")
    print("Message has been Modified or Incorrect MAC.")
```

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OUTPUT:-

```
17 received_message = input("Enter the Received Message: ").encode()
18 received_mac = input("Enter the received MAC: ")
19

PS C:\Users\mvlui> & C:\Users\mvlui\AppData\Local\Programs\Python\Python314\python.exe "d:/T079_VIBHUTI/CYBER SECURITY/T079_PRACT3.py"
Message: This is a confidential message
Generated MAC: ba495408ea1237f15e4b1c240421477faa90d5c63b8ec7770470b1e1ca899e
MAC Verification Successful: Message is authentic.
MAC Verification Failed: Message has been altered.
PS C:\Users\mvlui> & C:\Users\mvlui\AppData\Local\Programs\Python\Python314\python.exe "d:/T079_VIBHUTI/CYBER SECURITY/T079_PRACT3.py"
Enter the message: vibhuti

Generated MAC (HMAC-SHA256):
1e1ec10c1e10802d27ad67940db65664d8d3aabf3da1d34bbe95852ba12ea3

----- Verify Message -----
Enter the received message: vibhuti
Enter the received MAC: 1e1ec10c1e10802d27ad67940db65664d8d3aabf3da1d34bbe95852ba12ea3

Message Verified Successfully!
Integrity and Authenticity are Maintained.
PS C:\Users\mvlui> []
```

GUI

INPUT:-

```
import tkinter as tk
from tkinter import messagebox
import hmac
import hashlib

SECRET_KEY = b"mysecretkey"

def generate_mac():
    message = message_entry.get()

    if message == "":
        messagebox.showerror("Error", "Please enter a message.")
        return

    mac = hmac.new(SECRET_KEY, message.encode(),
hashlib.sha256).hexdigest()

    mac_entry.delete(0, tk.END)
    mac_entry.insert(0, mac)

def verify_mac():
    received_message = verify_message_entry.get()
```

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```
received_mac = verify_mac_entry.get()

if received_message == "" or received_mac == "":
    messagebox.showerror("Error", "Please fill all verification
fields.")
    return

new_mac = hmac.new(
    SECRET_KEY,
    received_message.encode(),
    hashlib.sha256
).hexdigest()

if hmac.compare_digest(received_mac, new_mac):
    messagebox.showinfo(
        "Verification",
        "Message Verified Successfully!\n\nIntegrity and Authenticity
are Maintained."
    )
else:
    messagebox.showerror(
        "Verification",
        "Verification Failed!\n\nMessage has been Modified or
Incorrect MAC."
    )

root = tk.Tk()
root.title("Message Authentication Code (MAC)")
root.geometry("600x400")
root.resizable(False, False)

tk.Label(root, text="Generate MAC", font=("Arial", 14,
"bold")).pack(pady=10)

tk.Label(root, text="Enter Message").pack()

message_entry = tk.Entry(root, width=60)
message_entry.pack()
```

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```
tk.Button(root, text="Generate MAC", command=generate_mac, bg="green",
fg="white").pack(pady=10)

tk.Label(root, text="Generated MAC").pack()

mac_entry = tk.Entry(root, width=70)
mac_entry.pack()

tk.Label(root,
text="-----").pack(pady=10)

tk.Label(root, text="Verify MAC", font=("Arial", 14, "bold")).pack()

tk.Label(root, text="Received Message").pack()

verify_message_entry = tk.Entry(root, width=60)
verify_message_entry.pack()

tk.Label(root, text="Received MAC").pack()

verify_mac_entry = tk.Entry(root, width=70)
verify_mac_entry.pack()

tk.Button(root, text="Verify", command=verify_mac, bg="blue",
fg="white").pack(pady=15)

root.mainloop()
```

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OUTPUT:-

